

Lecture 1b: Time Value of Money & Net Present Value

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- **The Time Value of Money (TVM)** is the change in the buying power of money over time. Money, and more specifically its value, is not conserved like mass, momentum, or energy. We capture this change using the concept of *discounting*.
- **Abstract assets** are sequences of current and future cash flows demarcated in a currency. This mental model can describe the financial value of anything you consider an asset: cash, stocks, bonds, or physical assets such as a car or a house.
- **The Net Present Value (NPV)** is the sum of all current and future cash flows discounted to the present. A positive NPV indicates that an investment is expected to generate more value than a benchmark risk-free alternative; a negative NPV indicates the opposite.

1 Introduction

Today, we introduce two ideas that we will return to over and over again throughout the course: the time value of money and the abstract asset. The time value of money is the principle that a dollar today has greater purchasing power than the same dollar received in the future, an observation with a long history in economics, going back at least to Fisher (1). The abstract asset is a mental model that represents anything of financial value as a sequence of cash flows occurring over time. Putting these two ideas together gives us our first practical decision-making tool, the net present value (NPV), which is the standard starting point for the valuation of projects, products, and investments (2). We begin with a question: *why* does money lose value over time?

2 The Time Value of Money

Money loses value over time for several reinforcing reasons. Inflation eats away purchasing power; for example, a 3% inflation rate means that 100 USD today has approximately 97 USD of buying power next year. Opportunity cost means that money in hand today can be invested to earn a return, so a dollar received in the future arrives having missed that opportunity. Risk means that promised future payments may not happen at all (a bird in the hand versus two in the bush). Finally, impatience: people simply prefer money now, a psychological preference for immediate gratification. The bottom line: a dollar today is worth more than a dollar tomorrow. Always.

Let's capture this idea mathematically. Suppose we have P USD today (units: current USD). Then, given a *discount rate* $\bar{r}$ (assumed constant), the future value F of our money (units: future USD) is:

$$F = \underbrace{(1 + \bar{r})}_{\mathcal{D}_{1,0}(\bar{r}) > 1} P$$

The conversion factor between current dollars and future dollars is called the discount factor.

Definition 1 (Discount Factor). *The discount factor $\mathcal{D}_{\star,0}(\bar{r})$ is the conversion factor between current dollars (period 0) and future dollars (period $\star$). Multiplying a present value by $\mathcal{D}_{\star,0}$ converts it to a future value; multiplying a future value by $\mathcal{D}_{\star,0}^{-1}$ converts it to a present value.*

The **discount rate** $\bar{r}$ goes by many names depending upon the context. It could be the interest rate we earn in a savings account, or the weighted average cost of capital, i.e., the rate at which a firm can borrow. We'll keep it general and call it the discount rate.

2.1 Multi-Period Discrete Discounting

When an investment lasts many periods, e.g., many years, we extend the present and future value relationship by compounding. At the end of each period we have a compounding event: all accrued interest is fully reinvested, i.e., added to the principal, thereby increasing the base on which future interest is calculated.

Assume discrete compounding with **one compounding event** per period (typical unit of a period: years). Let $\dot{c}_k$ denote the cash flow event in period $k = 0, 1, 2, \dots$, and let $r_{j+1,j} \geq 0$ denote the discount rate between periods $j \rightarrow (j+1)$. Then, by induction, cash flow events are related by the expression:

$$\dot{c}_k = \underbrace{\left[\prod_{j=0}^{k-1} (1 + r_{j+1,j}) \right]}_{\mathcal{D}_{k,0}(r) \geq 1} \cdot \dot{c}_0 \quad k = 0, 1, 2, \dots$$

where $\dot{c}_0$ is the cash flow event now (the present value) and $\dot{c}_k$ is the cash flow event in period k (the future value). The product term is the *multi-period discrete discount factor* $\mathcal{D}_{k,0}(r) \geq 1$. For $k = 0$ (now), the present and future values are equal, which means $\mathcal{D}_{0,0}(r) = 1$. For $k > 0$ (the future), the discount factor $\mathcal{D}_{k,0}(r) > 1$ for positive discount rates. Later in the course, we will see that there is information buried in the $r_{j+1,j}$ terms, the so-called **short rates** between periods, but often we assume the discount rate is constant in time (3).

Definition 2 (Constant discount rate). *Suppose we have a single compounding event per period, and the discount rate is constant, $\bar{r} \equiv r_{j+1,j}, \forall j$. Then the discrete multistep discount factor is $\mathcal{D}_{k,0}(\bar{r}) = (1 + \bar{r})^k$, where the period index k is typically measured in years.*

2.2 Multiple Compounding Events per Year

What if compounding occurs more than once per year, e.g., semi-annual (two events per year) or quarterly (four events per year) compounding? Suppose we have a **constant** annualized discount rate $\bar{r} \geq 0$ and n compounding events yearly. Then we can write the lifetime of the investment in terms of a duration $T \geq 0$ (units: years) instead of the period count $k \geq 0$, where $k = n \cdot T$, and the discrete discount factor becomes:

$$\mathcal{D}_{T,0}(\bar{r}) = (1 + \bar{r}/n)^{nT}$$

Example 1 (Semi-annual compounding). *Suppose we have semi-annual compounding, i.e., two events per year, so $n = 2$. Then for $k = 1$ the investment lifetime is $T = k/n = 1/2 = 0.5$ years; for $k = 2$, the lifetime is $T = 2/2 = 1$ year, and so on.*

2.3 Continuous Compounding

Now let's consider a different case. Suppose we let $n \rightarrow \infty$, i.e., we move to continuous compounding (a compounding event at every instant) for a fixed $T \geq 0$. What happens to the discount factor? For a

time-dependent discount rate $r(t)$, the continuous discount factor is:

$$\begin{aligned}\mathcal{D}_{T,0}(\bar{r}) &= \exp\left(\int_0^T r(u) du\right) \implies \text{assume } r \text{ constant} \\ &= \exp\left(\bar{r} \int_0^T 1 du\right) \\ &= \exp(\bar{r} \cdot T)\end{aligned}$$

Thus, the continuous discount factor is $\mathcal{D}_{T,0}(\bar{r}) = \exp(\bar{r} \cdot T)$. Consistent with the discrete case, $\mathcal{D}_{0,0}(\bar{r}) = 1$ for $T = 0$ (now), and $\mathcal{D}_{T,0}(\bar{r}) > 1$ for $T > 0$. In this course, we will use continuous discounting for equity and derivative instruments, while discrete discounting is the convention for United States Treasury securities.

How different are discrete and continuous compounding? Surprisingly close, and we can quantify the gap.

Proposition 1 (Compounding gap). *Let $\mathcal{D}_{T,0}(\bar{r}, n) = (1 + \bar{r}/n)^{nT}$ denote the discrete discount factor with n compounding events per year. Then, for large n , the relative gap between continuous and discrete compounding is:*

$$1 - \frac{\mathcal{D}_{T,0}(\bar{r}, n)}{e^{\bar{r}T}} \approx \frac{\bar{r}^2 T}{2n}$$

Proof. Write $(1 + \bar{r}/n)^{nT} = \exp[nT \ln(1 + \bar{r}/n)]$ and expand the logarithm, $\ln(1+x) = x - x^2/2 + x^3/3 - \dots$ with $x = \bar{r}/n$, to obtain $\mathcal{D}_{T,0}(\bar{r}, n) = e^{\bar{r}T} \exp\left(-\frac{\bar{r}^2 T}{2n} + \frac{\bar{r}^3 T}{3n^2} - \dots\right)$. Retaining the leading correction term and using $e^{-x} \approx 1 - x$ for small x gives the result. $\square$

For example, with $\bar{r} = 10\%$ and $T = 1$ year, moving from annual ($n = 1$) to daily ($n = 365$) compounding closes the gap to continuous compounding to roughly $\bar{r}^2/(2 \cdot 365) \approx 0.0014\%$. The practical message: as n grows, discrete compounding approaches the continuous limit **very** fast.

3 Who Sets the Discount Rate?

The discount rate is not a fixed universal constant; it is determined by the market, the industry, and the riskiness of the investment. Thus, an entirely unsatisfying but true answer is: *it depends*. However, let's dig a little deeper.

The United States Federal Reserve, created by the Federal Reserve Act of 1913, is the nation's central banking system, comprising a Board of Governors and twelve regional Federal Reserve Banks. The Federal Reserve conducts monetary policy, via tools such as open market operations, the discount rate, and reserve requirements, to promote maximum employment, stable prices, and moderate long-term interest rates. Within the Federal Reserve, the **Federal Open Market Committee (FOMC)** sets the federal funds rate, the interest rate at which banks borrow from one another overnight. Many other rates are pegged to the federal funds rate; FOMC decisions therefore ripple through the economy, affecting everything from mortgage rates to corporate borrowing costs, as well as consumer and investor sentiment.

What discount rate should *you* use in a calculation? It depends on risk: the higher the risk, the higher the required return (4). Rough ranges to calibrate your intuition:

- **Lower risk** (2–5%): Treasury bills, AAA corporate bonds. Sleep-well-at-night territory, although these instruments carry their own unique risks.

- **Moderate risk** (5 – 12%): investment-grade bonds (5 – 7%), large-capitalization stocks (8 – 12%), real estate (7 – 10%).
- **Higher risk** (12%+): small-capitalization stocks (12 – 18%), private equity (15 – 25%), startups (25 – 50%), cryptocurrencies (20 – 100%+, good luck!).

The key principle: investors demand compensation for bearing risk. There is no free lunch; if we accept more risk, we should expect (require) higher returns.

4 Abstract Assets and the Net Present Value

Definition 3 (Abstract asset). *An abstract asset is a sequence of current and future cash flows demarcated in a currency, e.g., Dollars, Euros, Yuan, or a cryptocurrency such as Bitcoin.*

The abstract asset is a useful mental model for describing the financial value of a process, product, or investment over time. However, there is a catch: money today is not worth the same as money tomorrow (purchasing power is not conserved). Thus, we must account for the time value of money when evaluating an abstract asset.

Suppose at each time period t we have a cash flow vector $\dot{\mathbf{c}}_t \in \mathbb{R}_{\geq 0}^n$ and a direction vector $\nu_t \in \mathbb{Z}_{\pm 1}^n$: if $\nu_{t,i} > 0$, cash flow event $\dot{c}_{t,i}$ is an inflow (money in our pocket); if $\nu_{t,i} < 0$, it is an outflow (money out of our pocket). The **net cash flow** during period t is the scalar product:

$$\bar{c}_t = \langle \dot{\mathbf{c}}_t, \nu_t \rangle = \sum_{i=1}^n \dot{c}_{t,i} \nu_{t,i}$$

Suppose our abstract asset has a life of $t = 0, 1, \dots, T$ periods. A naive (and incorrect) approach to valuing the asset is to simply sum the net cash flows, $\text{value} = \sum_{t=0}^T \bar{c}_t$. But remember: $\bar{c}_t$ for $t > 0$ has units of *future* dollars, so $\bar{c}_i$ and $\bar{c}_j$ (for $i \neq j$) are not directly comparable; they have different units. Discounting each cash flow back to today before summing fixes the units problem, and this is precisely the net present value.

Definition 4 (Net Present Value). *Given net cash flows $\bar{\mathbf{c}} = \{\bar{c}_0, \bar{c}_1, \dots, \bar{c}_T\}$ and discount factors $\mathcal{D}_{t,0}(r)$, the net present value is the sum of the discounted net cash flows:*

$$NPV = \sum_{t=0}^T \mathcal{D}_{t,0}^{-1}(r) \bar{c}_t$$

When the discount rate is constant over time, the NPV calculation collapses to an inner product between the inverse discount factors and the net cash flows:

$$NPV_{T,0}(\bar{r}) = \sum_{t=0}^T \mathcal{D}_{t,0}^{-1}(\bar{r}) \bar{c}_t = \langle \mathcal{D}_{\star,0}^{-1}(\bar{r}), \bar{\mathbf{c}} \rangle$$

The sign of the NPV tells us whether the investment is expected to create or destroy value:

- **Positive:** projected earnings (in present dollars) exceed anticipated costs (also in present dollars); the investment is potentially worthwhile.
- **Negative:** the costs outweigh the benefits; the investment may not be a good idea.

- **Zero:** the investment breaks even. Alternatively, we can interpret $NPV = 0$ as the value returned by a **risk-free** alternative investment; this interpretation will become clearer when we study United States Treasury securities in the next lecture.

Example 2 (Is buying a Tesla Model S worth it?). *We model the purchase, ownership, and sale of a Tesla Model S as an abstract asset. At $t = 0$ we have a large outflow (the purchase price). During each intermediate period we have smaller net flows, e.g., insurance and operating payments net of any savings (such as avoided fuel costs). At the final period T we have an inflow (the resale value) plus that period's net savings. Choosing a discount rate $\bar{r}$ and a compounding convention, we compute the discount factors $\mathcal{D}_{t,0}(\bar{r})$ for each period, invert them, and take the inner product with the net cash flows to obtain the NPV. For realistic parameter choices, the NPV is negative: a car is a cost, not an investment. The interesting questions are quantitative: how negative, and which assumptions (holding period, discount rate, resale value, avoided costs) move the answer the most? The complete, executable version of this example, including all parameter values and discussion questions, lives in the L1b worked-example notebook.*

5 Summary and Conclusions

In this lecture, we introduced the time value of money: money is not conserved, and the discount factor $\mathcal{D}_{t,0}$ is the machinery that converts cash flows between time periods so they can be compared in common units. We developed discount factors for discrete compounding (one or many compounding events per year) and for continuous compounding, and we showed that the two conventions converge quickly as the compounding frequency grows. We then introduced the abstract asset, a sequence of net cash flows, and the net present value, the inner product between inverse discount factors and net cash flows. The NPV converts the fuzzy question “is this a good deal?” into a number with a sign. Next time, we will apply this machinery to its most important special case: United States Treasury securities, where the cash flow schedule is known in advance and the discounting conventions are set by the market.

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